WORKED SOLUTION

STEP II 2011, Question 4

Pure • Trigonometric Equations

STEP II 2011, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q4

The key idea

Write $\cos\theta$ as the sine (or cosine) of a related angle, so that "$\cos\theta = \sin 4\theta$" becomes an equation between two cosines, which we can solve exactly. Expanding $\sin 4\theta$ in powers of $\sin\theta$ gives a cubic whose roots are the sines of those angles, and that is how a surd like $\sin 18^\circ$ appears. Part (ii) is a quadratic in $\sin^2 x$, and part (iii) is part (ii) in disguise once you spot an angle whose sine squared is $\frac{1}{4}$.

Solution

Part (i)

To compare like with like, write $\sin 4\theta = \cos(90^\circ - 4\theta)$. Two angles have the same cosine exactly when they are equal or opposite, up to multiples of 360°, so

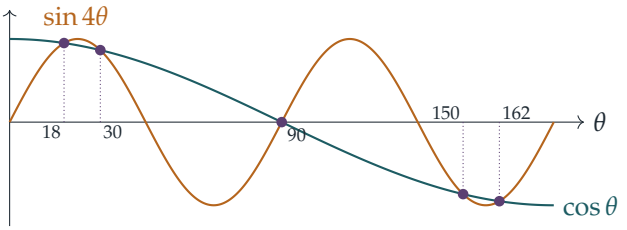
$$\cos\theta = \cos(90^\circ - 4\theta) \iff \theta = \pm(90^\circ - 4\theta) + 360^\circ k \quad \text{for some integer } k.$$

- With the $+$ sign: $5\theta = 90^\circ + 360^\circ k$, so $\theta = 18^\circ + 72^\circ k$. In $0^\circ < \theta < 180^\circ$ this gives $\theta = 18^\circ,\ 90^\circ,\ 162^\circ$.
- With the $-$ sign: $3\theta = 90^\circ - 360^\circ k$, so $\theta = 30^\circ - 120^\circ k$, giving $\theta = 30^\circ,\ 150^\circ$.

So the solutions are

$$\theta = 18^\circ,\ 30^\circ,\ 90^\circ,\ 150^\circ,\ 162^\circ.$$

The graphs below confirm there are exactly five crossings.



Now for the "hence". Write $s = \sin\theta$. Using the double-angle formulae twice,

$$\sin 4\theta = 2\sin 2\theta\cos 2\theta = 4\sin\theta\cos\theta\,(1 - 2\sin^2\theta),$$

so the equation $\cos\theta = \sin 4\theta$ becomes

$$\cos\theta\big(1 - 4s(1 - 2s^2)\big) = 0.$$

Ask yourself Why can we cancel the $\cos\theta$?

Of our five solutions, only $\theta = 90^\circ$ has $\cos\theta = 0$. The other four, $18^\circ, 30^\circ, 150^\circ, 162^\circ$, all have $\cos\theta \neq 0$, so for them we may divide by $\cos\theta$ and get

$$8s^3 - 4s + 1 = 0.$$

We know $s = \sin 30^\circ = \frac{1}{2}$ is one root, and factorising,

$$8s^3 - 4s + 1 = (2s - 1)(4s^2 + 2s - 1).$$

Now $\sin 18^\circ$ is a root of the cubic, and $0 < \sin 18^\circ < \sin 30^\circ = \frac{1}{2}$ because sine is increasing on $[0^\circ, 90^\circ]$. So $\sin 18^\circ \neq \frac{1}{2}$, and it must be a root of $4s^2 + 2s - 1 = 0$, whose roots are

$$s = \frac{-2 \pm \sqrt{4 + 16}}{8} = \frac{-1 \pm \sqrt{5}}{4}.$$

Only the $+$ root is positive, so

$$\sin 18^\circ = \tfrac{1}{4}(\sqrt{5} - 1),$$

as required.

Part (ii)

Let $s = \sin x$. Since $\sin^2 2x = 4\sin^2 x\cos^2 x = 4s^2(1 - s^2)$, the equation depends only on s:

$$4s^2 + 1 = 16s^2(1 - s^2) \iff 16s^4 - 12s^2 + 1 = 0.$$

This is a quadratic in s^2:

$$s^2 = \frac{12 \pm \sqrt{144 - 64}}{32} = \frac{12 \pm 4\sqrt{5}}{32} = \frac{3 \pm \sqrt{5}}{8}.$$

To take square roots, notice that

$$\left(\frac{1 + \sqrt{5}}{4}\right)^2 = \frac{6 + 2\sqrt{5}}{16} = \frac{3 + \sqrt{5}}{8}, \qquad \left(\frac{\sqrt{5} - 1}{4}\right)^2 = \frac{6 - 2\sqrt{5}}{16} = \frac{3 - \sqrt{5}}{8}.$$

Both values of s^2 lie between 0 and 1, so every one of the four square roots is a genuine value of $\sin x$, and each satisfies the original equation because the steps above were all reversible. The possible values are

$$\sin x = \tfrac{1}{4} + \tfrac{1}{4}\sqrt{5}, \qquad -\tfrac{1}{4} - \tfrac{1}{4}\sqrt{5}, \qquad -\tfrac{1}{4} + \tfrac{1}{4}\sqrt{5}, \qquad \tfrac{1}{4} - \tfrac{1}{4}\sqrt{5}.$$

By part (i), $\frac{1}{4}(\sqrt{5} - 1) = \sin 18^\circ$; and $\frac{1}{4}(1 + \sqrt{5}) = \sin 54^\circ$ (it is $\cos 36^\circ = 1 - 2\sin^2 18^\circ$, which you can check comes to the same thing).

Part (iii)

Divide the equation of part (ii) by 4:

$$\sin^2 x + \tfrac{1}{4} = \sin^2 2x.$$

Compare this with $\sin^2 3\alpha + \sin^2 5\alpha = \sin^2 6\alpha$. If we put $x = 3\alpha$, then $2x = 6\alpha$ automatically, and the two equations match provided

$$\sin^2 5\alpha = \tfrac{1}{4} \qquad \text{and} \qquad \sin 3\alpha \text{ is one of the four values in part (ii).}$$

Ask yourself Is it enough for $\sin 3\alpha$ to be one of those values?

Yes: in part (ii) we saw that the equation $\sin^2 x + \frac{1}{4} = \sin^2 2x$ involves x only through $s = \sin x$, and those four values of s satisfy it. So any x with one of those sines works.

Take $\alpha = 6°$. Then $5\alpha = 30°$, so $\sin^2 5\alpha = \frac{1}{4}$, and $3\alpha = 18°$, so $\sin 3\alpha = \frac{1}{4}(\sqrt{5} - 1)$, one of our values. Hence

$$\sin^2 18° + \sin^2 30° = \sin^2 36°,$$

and $\alpha = 6°$ is a solution.

Take $\alpha = 42°$. Then $5\alpha = 210°$, so $\sin 5\alpha = -\frac{1}{2}$ and $\sin^2 5\alpha = \frac{1}{4}$; and $3\alpha = 126°$, so $\sin 3\alpha = \sin 54° = \frac{1}{4}(1 + \sqrt{5})$, again one of our values. So $\alpha = 42°$ is a solution.

So two suitable values are $\alpha = 6°$ and $\alpha = 42°$.

The same method also gives $\alpha = 66°$ ($5\alpha = 330°$, $\sin 3\alpha = \sin 198° = -\sin 18°$) and $\alpha = 78°$ ($5\alpha = 390°$, $\sin 3\alpha = \sin 234° = -\sin 54°$). The other angle in range with $\sin^2 5\alpha = \frac{1}{4}$ is $\alpha = 30°$, and it fails, since $\sin 90° = 1$ is not one of our values. (Indeed $1 + \frac{1}{4} \neq \sin^2 180° = 0$.)

Checks

- $\sin 18° \approx 0.30902$ and $\frac{1}{4}(\sqrt{5} - 1) \approx 0.30902$.
- For $\alpha = 6°$: $\sin^2 18° + \sin^2 30° \approx 0.09549 + 0.25 = 0.34549$, and $\sin^2 36° \approx 0.34549$.
- A numerical scan of $\sin^2 3\alpha + \sin^2 5\alpha - \sin^2 6\alpha$ on $(0°, 90°)$ finds zeros at exactly $6°, 42°, 45°, 66°, 78°$. The extra one, $45°$, does not come from part (ii): there $\frac{1}{2} + \frac{1}{2} = 1$.

Common slips

- Forgetting the second family of solutions of $\cos A = \cos B$ (namely $A = -B + 360°k$), and so missing $30°$ and $150°$ in part (i).
- Cancelling $\cos\theta$ without comment. It is fine here, but only because the solution with $\cos\theta = 0$ is $90°$, not $18°$.
- Giving only the positive values of $\sin x$ in part (ii). All four are possible.

Beyond STEP

The exact value of $\sin 18^\circ$ comes from the fact that 18° is a fifth of a right angle, and so it is tied to the regular pentagon and the golden ratio: $\frac{1}{4}(1+\sqrt{5}) = \frac{1}{2}\varphi$. More generally, $\cos(360^\circ/n)$ can be written using square roots alone exactly when a regular n-gon can be constructed with ruler and compasses, which Gauss showed depends on the Fermat primes dividing n. This is a classic first application of field theory and Galois theory. Allenby's *Rings, Fields and Groups* builds up the algebra, and Needham's *Visual Complex Analysis* shows why roots of unity and polygons belong together.

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